

Ex. No. : 7

IPC USING SHARED MEMORY

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Date: 19.02.2025

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Aim:

To write a C program to implement Inter Process Communication (IPC) using shared memory between sender and receiver processes.

Algorithm:

Sender Process

1. Set the size of the shared memory segment.
2. Allocate the shared memory segment using `shmget()`.
3. Attach the shared memory segment using `shmat()`.
4. Write a string to the shared memory segment using `sprintf()`.
5. Set delay using `sleep()`.
6. Detach shared memory segment using `shmdt()`.

Receiver Process

1. Set the size of the shared memory segment.
2. Allocate the shared memory segment using `shmget()`.
3. Attach the shared memory segment using `shmat()`.
4. Print the shared memory contents sent by the sender process.
5. Detach shared memory segment using `shmdt()`.

Program Code: sender.c

```
#include <stdio.h>

#include <sys/ipc.h>

#include <sys/shm.h>

#include <unistd.h>

#include <string.h>
```

```

int main() {    key_t key = ftok("shmfile",65); // Generate unique key    int
shmidx = shmget(key, 1024, 0666|IPC_CREAT); // Create shared memory
char *str = (char*) shmat(shmidx, (void*)0, 0); // Attach to shared memory

    sprintf(str, "Welcome to Shared Memory");
printf("Message Sent: %s\n", str);

    sleep(5); // Delay to allow receiver to read
shmidx(str); // Detach from shared memory

    return 0;
}

```

receiver.c

```

#include <stdio.h>
#include <sys/ipc.h>
#include <sys/shm.h>
#include <unistd.h>

int main() {    key_t key = ftok("shmfile",65); // Generate same key    int
shmidx = shmget(key, 1024, 0666|IPC_CREAT); // Access shared memory
char *str = (char*) shmat(shmidx, (void*)0, 0); // Attach to shared memory

    printf("Message Received: %s\n", str);

    shmidx(str); // Detach from shared memory    shmctl(shmidx,
IPC_RMID, NULL); // Destroy the shared memory    return 0;

}

```

Sample Output:**Terminal 1:**

```
[root@localhost student]# gcc sender.c -o sender
```

```
[root@localhost student]# ./sender Message
```

Sent: Welcome to Shared Memory

Terminal 2:

```
[root@localhost student]# gcc receiver.c -o receiver
```

```
[root@localhost student]# ./receiver
```

Message Received: Welcome to Shared Memory

Result:

Thus, the C program for Inter Process Communication (IPC) using shared memory was successfully executed, and the message was successfully passed from the sender process to the receiver process.